

Computer Networks (CN) – Comprehensive Notes

1. OSI vs TCP/IP Models

(Already covered above, kept here for completeness)

OSI Model (7 Layers)

1. **Physical Layer** – Raw bit transmission, cables, voltages, physical connectors.
2. **Data Link Layer** – Error detection/correction, MAC addressing, framing.
3. **Network Layer** – Logical addressing, routing (IP).
4. **Transport Layer** – End-to-end delivery, segmentation, error control (TCP, UDP).
5. **Session Layer** – Establish, maintain, terminate communication sessions.
6. **Presentation Layer** – Data translation, encryption, compression.
7. **Application Layer** – User-facing services (HTTP, SMTP, FTP).

TCP/IP Model (4 Layers)

1. **Network Interface** – Physical & data link functions.
 2. **Internet** – IP addressing, routing.
 3. **Transport** – TCP, UDP.
 4. **Application** – All application-level protocols.
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2. Routing Algorithms

- **Distance Vector Routing (Bellman-Ford)**
 - **Link-State Routing (Dijkstra)**
 - **Hybrid protocols:** EIGRP.
 - **Path Vector Protocol:** BGP (Border Gateway Protocol) for internet-scale routing.
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3. Common Protocols

- **TCP** – Reliable, ordered, error-checked.
- **UDP** – Faster, connectionless, no guarantee.
- **HTTP/HTTPS** – Web communication (HTTPS is HTTP over TLS).
- **FTP/SFTP** – File transfer protocols.
- **SMTP, POP3, IMAP** – Email protocols.
- **DNS** – Domain name resolution.
- **DHCP** – Dynamic IP allocation.
- **SNMP** – Network management.

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- **ICMP** – Error messages (ping, traceroute).
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4. Network Devices

- **Switch** – Layer 2, connects LAN devices using MAC addresses.
 - **Router** – Layer 3, connects different networks using IP.
 - **Gateway** – Protocol converter between dissimilar networks.
 - **Hub** – Simple device, broadcasts to all ports (obsolete in modern networks).
 - **Access Point (AP)** – Wireless connectivity.
 - **Firewall** – Security filtering.
 - **Load Balancer** – Distributes network traffic.
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5. Network Topologies

Network topology refers to how devices are arranged and connected.

1. **Bus Topology**
 - All devices connected to a single backbone cable.
 - Pros: Cheap, easy to set up.
 - Cons: Single point of failure, collisions.
 2. **Star Topology**
 - Devices connected to a central hub/switch.
 - Pros: Easy to manage, one link failure doesn't affect others.
 - Cons: Hub failure stops entire network.
 3. **Ring Topology**
 - Devices connected in a closed loop.
 - Pros: Equal access, predictable performance.
 - Cons: One failure affects the whole network.
 4. **Mesh Topology**
 - Every device connected to every other.
 - Pros: High redundancy, fault tolerance.
 - Cons: Expensive, complex.
 5. **Hybrid Topology**
 - Combination of multiple topologies.
 6. **Tree Topology**
 - Hierarchical arrangement (root switch → branches).
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6. Transmission Media

Wired:

- **Twisted Pair Cable** (UTP/STP) – Ethernet networks.
- **Coaxial Cable** – TV, broadband internet.
- **Optical Fiber** – High-speed, long-distance communication.

Wireless:

- **Radio Waves** – Wi-Fi, Bluetooth.
 - **Microwaves** – Satellite links, point-to-point communication.
 - **Infrared** – Short-range device communication.
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7. Types of Networks

- **PAN** (Personal Area Network) – Very small range (Bluetooth).
 - **LAN** (Local Area Network) – Small geographical area (office, home).
 - **MAN** (Metropolitan Area Network) – City-wide.
 - **WAN** (Wide Area Network) – Country/worldwide (Internet).
 - **VPN** (Virtual Private Network) – Secure private connection over public networks.
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8. Switching Techniques

- **Circuit Switching** – Dedicated path for the entire communication (like telephone calls).
 - **Packet Switching** – Data divided into packets (used in the Internet).
 - **Message Switching** – Store-and-forward entire messages (obsolete).
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9. Error Detection & Correction

- **Parity Check** – Simple single-bit error detection.
 - **Checksum** – Detects errors in transmitted data.
 - **CRC (Cyclic Redundancy Check)** – Stronger detection for frames.
 - **Hamming Code** – Error detection & correction.
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10. Multiplexing Techniques

- **TDM (Time Division Multiplexing)** – Time slots for each signal.
 - **FDM (Frequency Division Multiplexing)** – Different frequency bands.
 - **WDM (Wavelength Division Multiplexing)** – Optical fiber-based.
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11. Network Security Concepts

- **Firewalls** – Block unauthorized access.
 - **Encryption** – Securing data (AES, RSA).
 - **VPNs** – Encrypted tunneling.
 - **IDS/IPS** – Intrusion detection/prevention.
 - **Authentication** – Verifying identity (passwords, biometrics).
 - **Access Control Lists (ACLs)** – Restrict network traffic.
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12. Cloud & Modern Networking

- **SDN (Software Defined Networking)** – Network control via software.
 - **NFV (Network Functions Virtualization)** – Virtualized network services.
 - **IoT Networking** – Connecting smart devices.
 - **5G Networks** – High-speed, low-latency mobile communication.
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13. Important Networking Commands

- `ping` – Test connectivity.
- `tracert` / `tracert` – Show path packets take.
- `ipconfig` / `ifconfig` – Display network settings.
- `netstat` – Show active connections.
- `nslookup` / `dig` – DNS queries.